

Foundation (Jalapeno)

Q1. Solve the equation $x^2 + 5x + 6 = 0$ by factoring.

- (A) $x = -2, -3$
- (B) $x = -1, -6$
- (C) $x = 2, 3$
- (D) $x = 1, 6$
- (E) $x = -2, 3$

Q2. Factor the expression $x^2 + 2x - 15$.

- (A) $(x + 5)(x - 3)$
- (B) $(x - 5)(x + 3)$
- (C) $(x + 2)(x - 15)$
- (D) $(x - 2)(x + 15)$
- (E) $(x + 1)(x - 15)$

Q3. Solve the equation $x^2 - 7x + 12 = 0$ by factoring.

- (A) $x = 3, 4$
- (B) $x = 2, 6$
- (C) $x = 1, 12$
- (D) $x = 3, 12$
- (E) $x = 4, 12$

Q4. Factor the expression $x^2 - 4x - 21$.

- (A) $(x - 7)(x + 3)$
- (B) $(x + 7)(x - 3)$
- (C) $(x - 2)(x + 21)$
- (D) $(x + 2)(x - 21)$
- (E) $(x + 1)(x - 21)$

Q5. Solve the equation $x^2 + x - 6 = 0$ by factoring.

- (A) $x = -3, 2$
- (B) $x = -2, 3$
- (C) $x = 1, 6$
- (D) $x = -1, 6$
- (E) $x = -2, -3$

Q6. Factor the expression $x^2 + 4x + 4$.

- (A) $(x + 2)^2$
- (B) $(x - 2)^2$
- (C) $(x + 1)(x + 4)$
- (D) $(x - 1)(x - 4)$
- (E) $(x + 2)(x - 2)$

Q7. Solve the equation $x^2 - 9x + 20 = 0$ by factoring.

- (A) $x = 4, 5$
- (B) $x = 2, 10$
- (C) $x = 1, 20$
- (D) $x = 4, 20$
- (E) $x = 5, 20$

Q8. Factor the expression $x^2 - 3x - 10$.

- (A) $(x - 5)(x + 2)$
- (B) $(x + 5)(x - 2)$
- (C) $(x - 2)(x + 10)$
- (D) $(x + 2)(x - 10)$
- (E) $(x + 1)(x - 10)$

Open Ended Questions (FRQ)

Q9. Solve the equation $x^2 + 2x - 6 = 0$ by factoring and verify the solutions.

Q10. Factor the expression $x^2 - 2x - 15$ and solve the equation $x^2 - 2x - 15 = 0$.

Chef's Tips

- Remind students to check their work.
- Encourage students to use factoring to solve quadratic equations.
- Emphasize the importance of verifying solutions.